

STAT 516 - Homework 5

Instructions: Please submit your homework as a single PDF on Gradescope. After uploading, please make sure each page is assigned to the correct question before submitting. Show your work clearly.

1. **(8 points)** Let X and Y be continuous random variables with joint density function

$$f(x, y) = \begin{cases} kx^{-2}e^{-y/2}, & x > 1, y > 0, \\ 0, & \text{otherwise.} \end{cases}$$

Find the value of k .

2. **(8 points)** Let X and Y be discrete random variables with joint probability function

$$f(x, y) = \begin{cases} \frac{1}{4}, & x = 1, 2 \text{ and } y = 1, 2, \\ 0, & \text{otherwise.} \end{cases}$$

Let $U = X + Y$. Find the probability function of U .

3. **(8 points)** A policy pays 200 dollars for each lost workday and 100 dollars for each doctor visit. Let X be the number of lost workdays and Y be the number of doctor visits. The joint probability function of X and Y is given by

$X \backslash Y$	0	1	2
0	0.20	0.10	0.05
1	0.08	0.12	0.07
2	0.06	0.10	0.08
3	0.04	0.06	0.04

Calculate the probability that the policy pays at least 500 dollars.

4. **(10 points)** Let X and Y be continuous random variables with joint density function

$$f(x, y) = \begin{cases} \frac{1}{4}xy, & 0 < x < 4, 0 < y < 1, \\ 0, & \text{otherwise.} \end{cases}$$

Calculate $P\left(\frac{X}{4} < Y < \frac{X}{2}\right)$. (Hint: You may use $P(Y < \frac{X}{2}) - P(Y < \frac{X}{4})$)

5. **(9 points)** Let X be the portion of an insurance claim representing damage to a building, and let Y be the portion representing damage to the remaining property. The joint density function is

$$f(x, y) = \begin{cases} 12(1 - x - y)^2, & x > 0, y > 0, x + y < 1, \\ 0, & \text{otherwise.} \end{cases}$$

Calculate the probability that the portion of the claim representing damage to the building is less than 0.25.

6. **(9 points)** Let X and Y be random losses with joint density function

$$f(x, y) = 4e^{-2x-2y}, \quad x > 0, y > 0.$$

An insurance policy reimburses the amount $X + Y$. Calculate the probability that the reimbursement is less than 2.

7. **(10 points)** A device stops running when the first of two components fails. The lifetimes of the two components, in hours, have joint density function

$$f(x, y) = \frac{x + 2y}{12}, \quad 0 < x < 2, 0 < y < 2.$$

Calculate the probability that the device fails during its first hour of operation, i.e., $P(\min(X, Y) < 1)$. (Hint: use the complement rule)

8. **(10 points)** Let X and Y have joint density function

$$f(x, y) = \begin{cases} \frac{1}{6}, & 0 < x < 3, x < y < x + 2, \\ 0, & \text{otherwise.} \end{cases}$$

Determine the marginal density function of Y .

9. **(9 points)** Let X and Y be discrete random variables with joint probability function

$$f(x, y) = \begin{cases} \frac{x + y}{21}, & (x, y) = (1, 1), (1, 2), (2, 1), (2, 2), (3, 1), (3, 2), \\ 0, & \text{otherwise.} \end{cases}$$

Determine the marginal probability function of X .

10. **(10 points)** Let X and Y be continuous random variables with joint density function

$$f(x, y) = \begin{cases} \frac{2}{5}(x + y), & 0 < x < 1, x < y < 2, \\ 0, & \text{otherwise.} \end{cases}$$

Find the marginal density function of Y .

11. **(9 points)** Let X and Y have a uniform joint distribution over the region

$$\{(x, y) : x^2 + y^2 \leq 4, x > 0, y > 0\}.$$

Find the marginal density function of X .